Model Selection

keys: cross validation

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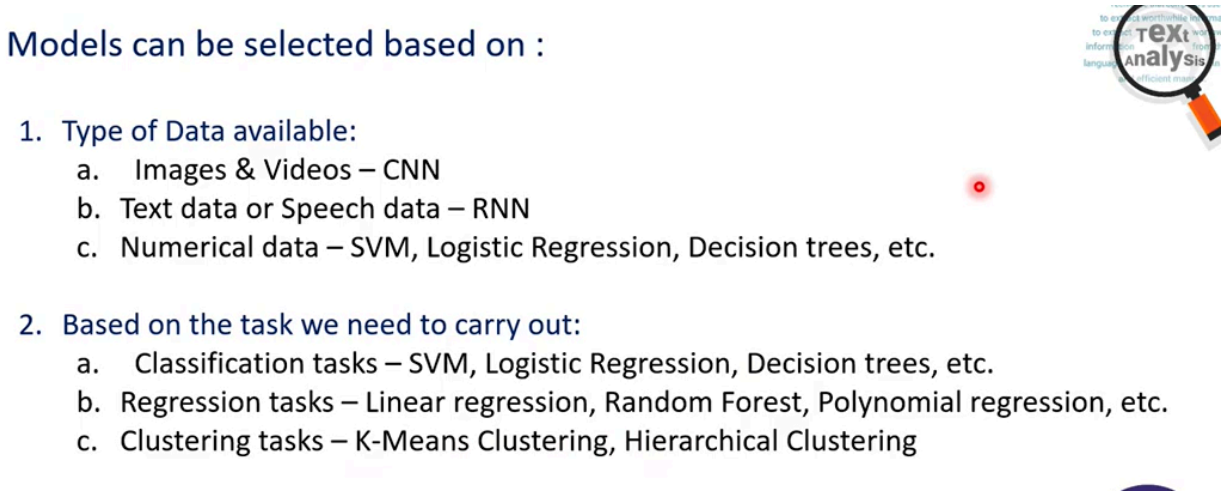
selecting a perfect model for best outcomes

factores : type of dataset,task,etc...



now while using simialar models like for ex:svm and logistic

we use CROSS VALIDATION funtion to pick the best model

	Dataset					Accuracy
Iteration 1	Train	Train	Train	Train	Test	88%
Iteration 2	Train	Train	Train	Test	Train	83%
Iteration 3	Train	Train	Test	Train	Train	86%
Iteration 4	Train	Test	Train	Train	Train	81%
Iteration 5	Test	Train	Train	Train	Train	84%

$$\text{Mean Accuracy} = \frac{88 + 83 + 86 + 81 + 84}{5} = 84.4\ \%$$

this funtion divides the data into subsets and train,tests reccurently by changing subsets and get accuracy and finally evaluates the mean accuracy

✓ *Accuracy score for SVM = 84.4 %*

✓ *Accuracy score for Logistic Regression = 88 %*

and picks the best model

```
from sklearn import datasets, linear_model
from sklearn.model_selection import cross_val_score
diabetes = datasets.load_diabetes()
X = diabetes.data[:150]
y = diabetes.target[:150]
```

```
lasso = linear_model.Lasso()
print(cross_val_score(lasso, X, y, cv=3))
```

cv=3 means it splits data set into 3 parts and 2/3 to train and 1/3 test on diff subsets